

# 10 TH MATHEMATICS IMPORTANT FORMULAE

## FULL CHAPTERS -2027

### CHAPTER 1: RELATIONS AND FUNCTIONS – POINTS TO REMEMBER

| S.NO | IMPORTANT FORMULAS / HINTS / POINTS                                                                                                                                                                                                | FORMULAE                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 1    | <b>Cartesian Product:</b><br>If $A$ and $B$ are two non-empty sets, then the set of all ordered pairs $(a, b)$ such that $a \in A, b \in B$ is called the Cartesian Product of $A$ and $B$ .                                       | $A \times B = \{(a, b) \mid a \in A, b \in B\}$                   |
| 2    | Cartesian product of $A$ and an empty set is an empty set.                                                                                                                                                                         | $A \times \phi = \phi$                                            |
| 3    | <b>Number of elements in Cartesian Product:</b><br>If $n(A) = p$ and $n(B) = q$ then the total number of elements in $A \times B$ is $pq$ .                                                                                        | $n(A \times B) = n(A) \times n(B) = pq$                           |
| 4    | <b>Relations:</b><br>A relation $R$ from a set $A$ to a set $B$ is a subset of the Cartesian product $A \times B$ satisfying some specified conditions.                                                                            | $R \subseteq A \times B$                                          |
| 5    | Total number of relations: If $n(A) = p, n(B) = q$ , then the total number of relations that exist from $A$ to $B$ is $2^{pq}$ .                                                                                                   | Total Relations = $2^{pq}$                                        |
| 6    | <b>Functions:</b><br>A relation $f$ between two non-empty sets $X$ and $Y$ is called a function from $X$ to $Y$ if, for each $x \in X$ there exists only one $y \in Y$ such that $(x, y) \in f$ .                                  | $f = \{(x, y) \mid \text{for all } x \in X, y \in Y\}$            |
| 7    | <b>Number of functions:</b><br>If $A$ and $B$ are finite sets such that $n(A) = p, n(B) = q$ then the total number of functions that exist from $A$ to $B$ is $q^p$ .                                                              | Total Functions = $q^p$                                           |
| 8    | <b>One-one Function (Injection):</b><br>A function $f: A \rightarrow B$ is called one-one if distinct elements of $A$ have distinct images in $B$ .                                                                                | If $f(a_1) = f(a_2)$ implies $a_1 = a_2$ for all $a_1, a_2 \in A$ |
| 9    | <b>Onto Function (Surjection):</b><br>A function $f: A \rightarrow B$ is an onto function if the range of $f$ is equal to the co-domain of $f$ .                                                                                   | Range of $f = B$                                                  |
| 10   | <b>Bijection:</b><br>If a function $f: A \rightarrow B$ is both one-one and onto, then $f$ is called a bijection from $A$ to $B$ .                                                                                                 | One-one and Onto                                                  |
| 11   | <b>Composition of Functions:</b><br>For three non-empty sets $A, B$ and $C$ , if $f: A \rightarrow B$ and $g: B \rightarrow C$ are two functions, then the composition of $f$ and $g$ is a function $g \circ f: A \rightarrow C$ . | $g \circ f(x) = g(f(x))$ for all $x \in A$                        |
| 12   | <b>Commutativity</b> of composition: In general, composition of functions is <b>not commutative</b> .                                                                                                                              | $f \circ g \neq g \circ f$                                        |

| S.NO | IMPORTANT FORMULAS / HINTS / POINTS                                                                             | FORMULAE                                    |
|------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 13   | <b>Associativity</b> of composition: Composition of functions is <b>always associative</b> .                    | $f \circ (g \circ h) = (f \circ g) \circ h$ |
| 14   | <b>Identity function:</b> A function $f: \mathbb{R} \rightarrow \mathbb{R}$ where every element maps to itself. | $f(x) = x$                                  |
| 15   | <b>Constant function:</b><br>A function $f: A \rightarrow B$ where the range contains only one element.         | $f(x) = c$                                  |
| 16   | <b>Linear function:</b><br>A polynomial function of degree 1. Geometric representation is a straight line.      | $f(x) = ax + b, a \neq 0$                   |
| 17   | <b>Quadratic function:</b><br>A polynomial function of degree 2. Geometric representation is a parabola.        | $f(x) = ax^2 + bx + c, a \neq 0$            |
| 18   | <b>Cubic function:</b> A polynomial function of degree 3.                                                       | $f(x) = ax^3 + bx^2 + cx + d, a \neq 0$     |
| 19   | <b>Reciprocal function:</b><br>A function that maps an element to its reciprocal.                               | $f(x) = \frac{1}{x}, x \neq 0$              |

## CHAPTER 2: NUMBERS AND SEQUENCES - POINTS TO REMEMBER

| S.No | Important Formulas / Hints / Points                                                                                                                                                                                                                       | Formulae                                                                                              |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1    | <b>Euclid's Division Lemma:</b><br>If $a$ and $b$ are two positive integers, then there exist unique integers $q$ and $r$ such that $a = bq + r$ , where $0 \leq r < b$ .                                                                                 | $a = bq + r, 0 \leq r < b$                                                                            |
| 2    | <b>Generalised Euclid's Division Lemma:</b><br>If $a$ and $b$ are ( $b \neq 0$ ) any two integers then there exist unique integers $q$ and $r$ such that $a = bq + r$ , where $0 \leq r <  b $ .                                                          | $a = bq + r, 0 \leq r <  b $                                                                          |
| 3    | <b>Euclid's Division Algorithm:</b><br>It provides an easier way to compute the Highest Common Factor (HCF) of two given positive integers. If $a = bq + r$ , then every common divisor of $a$ and $b$ is a common divisor of $b$ and $r$ and vice-versa. | $HCF(a, b) = HCF(b, r)$                                                                               |
| 4    | <b>Fundamental Theorem of Arithmetic:</b><br>Every positive integer (except the number 1) can be represented in exactly one way apart from rearrangement as a product of one or more primes.                                                              | $N = p_1^{q_1} \times p_2^{q_2} \times \dots \times p_n^{q_n}$                                        |
| 5    | <b>Modular Arithmetic:</b><br>Two integers $a$ and $b$ are congruent modulo $m$ , written as $a \equiv b \pmod{m}$ , if they leave the same remainder when divided by $m$ .                                                                               | $a \equiv b \pmod{m}$ implies<br>$a - b = km$                                                         |
| 6    | <b>Connecting Euclid's Lemma and Modular Arithmetic:</b> The equation $n = mq + r$ through Euclid's Division lemma can also be written as $n \equiv r \pmod{m}$ .                                                                                         | $n \equiv r \pmod{m}$                                                                                 |
| 7    | <b>Modulo Operations:</b> If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then addition, subtraction, and multiplication hold.                                                                                                                         | $(a + c) \equiv (b + d) \pmod{m}$<br>$(a - c) \equiv (b - d) \pmod{m}$<br>$(ac) \equiv (bd) \pmod{m}$ |
| 8    | <b>Sequence:</b><br>A real valued sequence is a function defined on the set of natural numbers and taking real values.                                                                                                                                    | $a_1, a_2, a_3, \dots, a_n, \dots$                                                                    |
| 9    | <b>Arithmetic Progression (A.P.):</b><br>An A.P. is a sequence whose successive terms differ by a constant number called the common difference ( $d$ ).                                                                                                   | $a, a + d, a + 2d, \dots$                                                                             |
| 10   | <b>General term of an A.P.:</b><br>The $n^{th}$ term of an A.P. with first term $a$ and common difference $d$ .                                                                                                                                           | $t_n = a + (n - 1)d$                                                                                  |
| 11   | <b>Number of terms in an A.P.:</b><br>If $l$ is the last term and $a$ is the first term, the number of terms $n$ is given by:                                                                                                                             | $n = \left( \frac{l - a}{d} \right) + 1$                                                              |

| S.No | Important Formulas / Hints / Points                                                                                                                                           | Formulae                                             |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 12   | <b>Sum to n terms of an A.P. (<math>S_n</math>):</b><br>The sum of the first $n$ terms of an A.P.                                                                             | $S_n = \frac{n}{2}[2a + (n - 1)d]$                   |
| 13   | <b>Sum to n terms of an A.P. (Alternative):</b><br>When the first term $a$ and last term $l$ are known.                                                                       | $S_n = \frac{n}{2}[a + l]$                           |
| 14   | <b>Condition for three numbers in A.P.:</b><br>Three non-zero numbers $a, b, c$ are in A.P. if and only if:                                                                   | $2b = a + c$                                         |
| 15   | <b>Geometric Progression (G.P.):</b><br>A G.P. is a sequence in which each term is obtained by multiplying a fixed non-zero number (common ratio $r$ ) to the preceding term. | $a, ar, ar^2, \dots, ar^{n-1}, \dots$                |
| 16   | <b>General term of a G.P.:</b><br>The $n^{th}$ term of a G.P. with first term $a$ and common ratio $r$ .                                                                      | $t_n = ar^{n-1}$                                     |
| 17   | <b>Sum to n terms of a G.P. (<math>r \neq 1</math>):</b>                                                                                                                      | $S_n = \frac{a(r^n - 1)}{r - 1}$                     |
| 18   | <b>Sum to n terms of a G.P. (<math>r = 1</math>):</b>                                                                                                                         | $S_n = na$                                           |
| 19   | <b>Sum to infinite terms of a G.P.:</b><br>The sum of infinite terms where $-1 < r < 1$ .                                                                                     | $S_\infty = \frac{a}{1 - r}$                         |
| 20   | <b>Condition for three numbers in G.P.:</b><br>Three non-zero numbers $a, b, c$ are in G.P. if and only if:                                                                   | $b^2 = ac$                                           |
| 21   | <b>Special Series (Sum of first n natural numbers):</b>                                                                                                                       | $\sum_{k=1}^n k = \frac{n(n+1)}{2}$                  |
| 22   | <b>Special Series (Sum of first n odd natural numbers):</b>                                                                                                                   | $\sum_{k=1}^n (2k - 1) = n^2$                        |
| 23   | <b>Special Series (Sum of squares of first n natural numbers):</b>                                                                                                            | $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$          |
| 24   | <b>Special Series (Sum of cubes of first n natural numbers):</b>                                                                                                              | $\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2$ |

## CHAPTER 3: ALGEBRA - POINTS TO REMEMBER

| S.No | Important Formulas / Hints / Points                                                                                                                                                                                                                              | Formulae                                                 |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 1    | <b>Linear Equation in Two Variables:</b><br>A first-degree equation in two variables $x$ and $y$ . It represents a straight line.                                                                                                                                | $ax + by + c = 0, a, b \neq 0$                           |
| 2    | <b>System of Linear Equations in Three Variables:</b><br>General form of linear equation in three variables. It represents a plane in three-dimensional space.                                                                                                   | $ax + by + cz + d = 0$                                   |
| 3    | <b>Nature of Solutions (Three Variables):</b> The system may have:<br>(i) Unique solution (intersect at one point)<br><br>(ii) Infinitely many solutions (planes coincide or intersect in a line)<br><br>(iii) No solution (do not intersect at a common point). | (i) Unique<br><br>(ii) Infinite<br><br>(iii) No Solution |
| 4    | <b>GCD of Polynomials:</b><br>The Greatest Common Divisor (GCD) of two or more polynomials is the polynomial of the highest degree that exactly divides each of the given polynomials.                                                                           | GCD                                                      |
| 5    | <b>LCM of Polynomials:</b><br>The Least Common Multiple (LCM) of two or more polynomials is the polynomial of the lowest degree that is exactly divisible by each of the given polynomials.                                                                      | LCM                                                      |
| 6    | <b>Relationship between LCM and GCD:</b><br>For any two polynomials $p(x)$ and $q(x)$ , the product of the polynomials is equal to the product of their LCM and GCD.                                                                                             | $p(x) \times q(x) = \text{LCM} \times \text{GCD}$        |
| 7    | <b>Rational Expression:</b><br>An expression of the form $\frac{p(x)}{q(x)}$ where $p(x)$ and $q(x)$ are polynomials and $q(x) \neq 0$ .                                                                                                                         | $\frac{p(x)}{q(x)}, q(x) \neq 0$                         |
| 8    | <b>Excluded Value:</b><br>A value of the variable that makes the denominator of a rational expression (in its lowest form) zero, causing the expression to be undefined.                                                                                         | Values $q(x) = 0$                                        |
| 9    | <b>Quadratic Polynomial:</b><br>A polynomial of degree two. It can have at most two zeroes. The graph is a parabola that intersects the $x$ -axis at its zeroes.                                                                                                 | $p(x) = ax^2 + bx + c, a \neq 0$                         |
| 10   | <b>Quadratic Equation:</b><br>An equation formed by equating a quadratic polynomial to zero.                                                                                                                                                                     | $ax^2 + bx + c = 0, a \neq 0$                            |

| S.No | Important Formulas / Hints / Points                                                                                                                                                                                             | Formulae                                                            |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 11   | <b>Roots by Quadratic Formula:</b><br>The roots of the quadratic equation $ax^2 + bx + c = 0$ can be found using the formula.                                                                                                   | $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$                            |
| 12   | <b>Sum of the Roots:</b><br>For the quadratic equation $ax^2 + bx + c = 0$ , if $\alpha$ and $\beta$ are the roots.                                                                                                             | $\alpha + \beta = -\frac{b}{a}$                                     |
| 13   | <b>Product of the Roots:</b><br>For the quadratic equation $ax^2 + bx + c = 0$ , if $\alpha$ and $\beta$ are the roots.                                                                                                         | $\alpha\beta = \frac{c}{a}$                                         |
| 14   | <b>Formation of Quadratic Equation:</b><br>If the roots $\alpha$ and $\beta$ are given, the equation can be formed.                                                                                                             | $x^2 - (\alpha + \beta)x + \alpha\beta = 0$                         |
| 15   | <b>Discriminant (<math>\Delta</math>):</b><br>The value under the square root in the quadratic formula determines the nature of the roots.                                                                                      | $\Delta = b^2 - 4ac$                                                |
| 16   | <b>Nature of Roots (Real and Unequal):</b><br>If the discriminant is strictly greater than zero.                                                                                                                                | $\Delta > 0$                                                        |
| 17   | <b>Nature of Roots (Real and Equal):</b> If the discriminant is equal to zero.                                                                                                                                                  | $\Delta = 0$                                                        |
| 18   | <b>Nature of Roots (No Real Roots):</b> If the discriminant is less than zero.                                                                                                                                                  | $\Delta < 0$                                                        |
| 19   | <b>Direct Variation:</b><br>Two quantities $x$ and $y$ are said to vary directly if they increase or decrease together such that their ratio is constant. The graph is a straight line passing through the origin.              | $y = kx, k > 0 \left(\frac{y}{x} = k\right)$                        |
| 20   | <b>Inverse Variation:</b><br>Two quantities $x$ and $y$ are said to vary inversely if an increase in one causes a proportional decrease in the other such that their product is constant. The graph is a rectangular hyperbola. | $xy = k, k > 0 \left(y = \frac{k}{x}\right)$                        |
| 21   | <b>Matrix:</b><br>A rectangular array of elements arranged in rows and columns.                                                                                                                                                 | $A = [a_{ij}]_{m \times n}$                                         |
| 22   | <b>Order of a Matrix:</b><br>If a matrix has $m$ rows and $n$ columns, its order is $m \times n$ (read as 'm by n').                                                                                                            | $m \times n$                                                        |
| 23   | <b>Transpose of a Matrix:</b><br>Obtained by interchanging rows into columns or columns into rows. Denoted by $A^T$ .                                                                                                           | $(A^T)^T = A$                                                       |
| 24   | <b>Matrix Addition/Subtraction:</b><br>Possible only if the two matrices have the same order. Operations are performed on corresponding elements.                                                                               | $A \pm B = [a_{ij} \pm b_{ij}]$                                     |
| 25   | <b>Matrix Multiplication:</b><br>Two matrices $A$ and $B$ can be multiplied to get $AB$ only if the number of columns in $A$ equals the number of rows in $B$ .                                                                 | If $A_{m \times p}$ and $B_{p \times n}$ , then $(AB)_{m \times n}$ |
| 26   | <b>Properties of Matrix Multiplication:</b><br>Matrix multiplication is NOT commutative in general.                                                                                                                             | $AB \neq BA \text{ (generally)}$                                    |

### FUNDAMENTAL ALGEBRAIC IDENTITIES

|   |                              |                                                     |
|---|------------------------------|-----------------------------------------------------|
| 1 | Square of a Sum & Difference | $(a \pm b)^2 = a^2 \pm 2ab + b^2$                   |
| 2 | Difference of Squares        | $a^2 - b^2 = (a + b)(a - b)$                        |
| 3 | Square of a Trinomial        | $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ |
| 4 | Cube of a Sum & Difference   | $(a \pm b)^3 = a^3 \pm b^3 \pm 3ab(a \pm b)$        |
| 5 | Sum & Difference of Cubes    | $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$         |
| 6 | Sum of Squares               | $a^2 + b^2 = (a + b)^2 - 2ab$                       |
| 7 | Difference                   | $a - b = \sqrt{(a + b)^2 - 4ab}$                    |

### FUNDAMENTAL ALGEBRAIC IDENTITIES

|   |                                             |                                                                          |
|---|---------------------------------------------|--------------------------------------------------------------------------|
| 1 | Sum and Product of Roots                    | $\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$         |
| 2 | Formation of Quadratic Eq.                  | $x^2 - (\alpha + \beta)x + \alpha\beta = 0$                              |
| 3 | $\alpha, \beta$ Identity: Sum of Squares    | $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$                 |
| 4 | $\alpha, \beta$ Identity: Difference        | $\alpha - \beta = \pm\sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$           |
| 5 | $\alpha, \beta$ Identity: Sum of Cubes      | $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$ |
| 6 | $\alpha, \beta$ Identity: Diff. of Squares  | $\alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta)$                  |
| 7 | $\alpha, \beta$ Identity: Sum of Power four | $\alpha^4 + \beta^4 = (\alpha + \beta)^4 - 4\alpha^2\beta^2$             |

## CHAPTER 4: GEOMETRY - POINTS TO REMEMBER

| S.No | Important Formulas / Hints / Points                                                                                                                                                             | Formulae                                                                              |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1    | <b>Similar Triangles:</b><br>Two triangles are similar if their corresponding angles are equal and their corresponding sides are in the same ratio or proportional.                             | $\triangle ABC \sim \triangle PQR$<br>$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP}$ |
| 2    | <b>AA Similarity Criterion:</b><br>If two angles of one triangle are respectively equal to two angles of another triangle, then the two triangles are similar.                                  | $\angle A = \angle P, \angle B = \angle Q$                                            |
| 3    | <b>SAS Similarity Criterion:</b><br>If one angle of a triangle is equal to one angle of another triangle and the sides including these angles are proportional, then the triangles are similar. | $\angle A = \angle P, \frac{AB}{PQ} = \frac{AC}{PR}$                                  |
| 4    | <b>SSS Similarity Criterion:</b><br>If three sides of a triangle are proportional to the corresponding sides of another triangle, then the two triangles are similar.                           | $\frac{AB}{PQ} = \frac{AC}{PR} = \frac{BC}{QR}$                                       |
| 5    | <b>Ratio of Perimeters:</b><br>If two triangles are similar, then the ratio of the corresponding sides is equal to the ratio of their corresponding perimeters.                                 | $\frac{AB}{PQ} = \frac{AB + BC + CA}{PQ + QR + RP}$                                   |
| 6    | <b>Ratio of Areas:</b><br>The ratio of the area of two similar triangles is equal to the ratio of the squares of their corresponding sides.                                                     | $\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle PQR)} = \frac{AB^2}{PQ^2}$   |
| 7    | <b>Basic Proportionality Theorem (Thales Theorem):</b><br>A straight line drawn parallel to a side of a triangle intersecting the other two sides, divides the sides in the same ratio.         | If $DE \parallel BC$ ,<br>$\frac{AD}{DB} = \frac{AE}{EC}$                             |
| 8    | <b>Angle Bisector Theorem:</b><br>The internal bisector of an angle of a triangle divides the opposite side internally in the ratio of the corresponding sides containing the angle.            | If $AD$ bisects $\angle A$ ,<br>$\frac{AB}{AC} = \frac{BD}{DC}$                       |
| 9    | <b>Pythagoras Theorem:</b><br>In a right angled triangle, the square on the hypotenuse is equal to the sum of the squares on the other two sides.                                               | In $\triangle ABC, \angle A = 90^\circ$<br>$BC^2 = AB^2 + AC^2$                       |
| 10   | <b>Converse of Pythagoras Theorem:</b><br>If the square of the longest side of a triangle is equal to sums of squares of other two sides, then the triangle is a right angle triangle.          | If $c^2 = a^2 + b^2, \angle C = 90^\circ$                                             |
| 11   | <b>Tangent to a Circle:</b><br>A tangent to a circle is perpendicular to the radius at the point of contact.                                                                                    | $OP \perp AB$                                                                         |
| 12   | <b>Tangents from an Exterior Point:</b><br>Two tangents can be drawn from an exterior point to a circle, and                                                                                    | $PA = PB$                                                                             |

| S.No | Important Formulas / Hints / Points                                                                                                                                                                                           | Formulae                                                       |
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|      | their lengths are equal.                                                                                                                                                                                                      |                                                                |
| 13   | <b>Alternate Segment Theorem:</b><br>If a line touches a circle and from the point of contact a chord is drawn, the angles between the tangent and the chord are equal to the angles in the corresponding alternate segments. | Angle between tangent and chord = Angle in alternate segment   |
| 14   | <b>Ceva's Theorem:</b><br>A necessary and sufficient condition for cevians $AD, BE, CF$ to be concurrent is that the product of the ratios is 1.                                                                              | $\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1$  |
| 15   | <b>Menelaus Theorem:</b><br>A necessary and sufficient condition for points $P, Q, R$ on the respective sides $BC, CA, AB$ of $\triangle ABC$ to be collinear is that the product of the directed segments is -1.             | $\frac{BP}{PC} \times \frac{CQ}{QA} \times \frac{AR}{RB} = -1$ |

## CHAPTER 5: COORDINATE GEOMETRY - POINTS TO REMEMBER

| S.No | Important Formulas / Hints / Points                                                                                                                                 | Formulae                                                                                                                                                                                                          |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Area of a Triangle:</b><br>The area of a triangle formed by the points $(x_1, y_1)$ , $(x_2, y_2)$ and $(x_3, y_3)$ .                                            | $\frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix}$ $\frac{1}{2}  (x_1y_2 + x_2y_3 + x_3y_1) - (x_2y_1 + x_3y_2 + x_1y_3)  \text{ sq.units}$                               |
| 2    | <b>Collinearity of Three Points:</b><br>Three points $A(x_1, y_1)$ , $B(x_2, y_2)$ and $C(x_3, y_3)$ are collinear if and only if the area of $\triangle ABC = 0$ . | Area of $\triangle ABC = 0$<br>$\frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix} = 0$ $x_1y_2 + x_2y_3 + x_3y_1 = x_2y_1 + x_3y_2 + x_1y_3$                               |
| 3    | <b>Collinearity by Slope:</b><br>Three points are collinear if the slope of any two pairs of points are equal.                                                      | Slope of AB = Slope of BC                                                                                                                                                                                         |
| 4    | <b>Area of a Quadrilateral:</b><br>The area of a quadrilateral formed by the four points $(x_1, y_1)$ , $(x_2, y_2)$ , $(x_3, y_3)$ and $(x_4, y_4)$ .              | $\frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_4 & x_1 \\ y_1 & y_2 & y_3 & y_4 & y_1 \end{vmatrix}$ $\frac{1}{2}  (x_1y_2 + x_2y_3 + x_3y_4 + x_4y_1) - (x_2y_1 + x_3y_2 + x_4y_3 + x_1y_4)  \text{ sq.units}$ |
| 5    | <b>Slope of a Line (Angle given):</b><br>If a line makes an angle $\theta$ with the positive direction of X-axis.                                                   | $m = \tan\theta$                                                                                                                                                                                                  |
| 6    | <b>Slope of a Line (Two points given):</b><br>If $A(x_1, y_1)$ and $B(x_2, y_2)$ are two distinct points.                                                           | $m = \frac{y_2 - y_1}{x_2 - x_1}$                                                                                                                                                                                 |
| 7    | <b>Slope of a Line (Equation given):</b><br>For a general equation of a straight line $ax + by + c = 0$ .                                                           | $m = -\frac{a}{b} \left( m = -\frac{\text{coefficient of } x}{\text{coefficient of } y} \right)$                                                                                                                  |
| 8    | <b>Parallel Lines:</b><br>Two straight lines are parallel if and only if their slopes are equal.                                                                    | $m_1 = m_2$                                                                                                                                                                                                       |
| 9    | <b>Perpendicular Lines:</b><br>Two straight lines with well-defined slopes $m_1$ and $m_2$ are perpendicular if and only if their product is -1.                    | $m_1 \times m_2 = -1$                                                                                                                                                                                             |
| 10   | <b>General Form of Straight Line:</b><br>The standard form of a linear equation in two variables representing a straight line.                                      | $ax + by + c = 0$                                                                                                                                                                                                 |
| 11   | <b>Intercept Form:</b><br>Equation of a line making intercepts $a$ and $b$ on the $x$ and $y$ axes respectively.                                                    | $\frac{x}{a} + \frac{y}{b} = 1$                                                                                                                                                                                   |

| S.No | Important Formulas / Hints / Points                                                                     | Formulae                                                   |
|------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 12   | <b>Point-Slope Form:</b><br>Equation of a line passing through a point $(x_1, y_1)$ with slope $m$ .    | $y - y_1 = m(x - x_1)$                                     |
| 13   | <b>Slope-Intercept Form:</b><br>Equation of a line with slope $m$ and $y$ -intercept $c$ .              | $y = mx + c$                                               |
| 14   | <b>Equation of lines parallel to axes:</b>                                                              | Parallel to X-axis: $y = b$<br>Parallel to Y-axis: $x = c$ |
| 15   | <b>Two Point Form:</b><br>Equation of a line passing through two points $(x_1, y_1)$ and $(x_2, y_2)$ . | $\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$    |

## CHAPTER 6: TRIGONOMETRY - POINTS TO REMEMBER

| S.No | Important Formulas / Hints / Points                                                                                       | Formulae                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 1    | <b>Sine Ratio:</b><br>In a right-angled triangle, it's the ratio of the length of the opposite side to the hypotenuse.    | $\sin\theta = \frac{\text{Opposite side}}{\text{Hypotenuse}}$                                    |
| 2    | <b>Cosine Ratio:</b><br>In a right-angled triangle, it's the ratio of the length of the adjacent side to the hypotenuse.  | $\cos\theta = \frac{\text{Adjacent side}}{\text{Hypotenuse}}$                                    |
| 3    | <b>Tangent Ratio:</b><br>It's the ratio of sine to cosine, or opposite side to adjacent side.                             | $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\text{Opposite side}}{\text{Adjacent side}}$ |
| 4    | <b>Cosecant Ratio:</b> The reciprocal of sine.                                                                            | $\csc\theta = \frac{1}{\sin\theta}$                                                              |
| 5    | <b>Secant Ratio:</b> The reciprocal of cosine.                                                                            | $\sec\theta = \frac{1}{\cos\theta}$                                                              |
| 6    | <b>Cotangent Ratio:</b> The reciprocal of tangent, or cosine to sine.                                                     | $\cot\theta = \frac{1}{\tan\theta} = \frac{\cos\theta}{\sin\theta}$                              |
| 7    | <b>Complementary Angles (Sine/Cosine):</b> The sine of an angle is equal to the cosine of its complement, and vice versa. | $\sin(90^\circ - \theta) = \cos\theta$<br>$\cos(90^\circ - \theta) = \sin\theta$                 |
| 8    | <b>Complementary Angles (Tangent/Cotangent):</b> The tangent of an angle is equal to the cotangent of its complement.     | $\tan(90^\circ - \theta) = \cot\theta$<br>$\cot(90^\circ - \theta) = \tan\theta$                 |
| 9    | <b>Complementary Angles (Cosecant/Secant):</b> The cosecant of an angle is equal to the secant of its complement.         | $\csc(90^\circ - \theta) = \sec\theta$<br>$\sec(90^\circ - \theta) = \csc\theta$                 |
| 10   | <b>Fundamental Identity 1:</b><br>The sum of the squares of sine and cosine is always 1.                                  | $\sin^2\theta + \cos^2\theta = 1$                                                                |
| 11   | <b>Derived Identity 1a:</b><br>Expressing sine squared in terms of cosine squared.                                        | $\sin^2\theta = 1 - \cos^2\theta$                                                                |
| 12   | <b>Derived Identity 1b:</b><br>Expressing cosine squared in terms of sine squared.                                        | $\cos^2\theta = 1 - \sin^2\theta$                                                                |
| 13   | <b>Fundamental Identity 2:</b><br>The relationship between tangent and secant.                                            | $1 + \tan^2\theta = \sec^2\theta$                                                                |
| 14   | <b>Derived Identity 2a:</b><br>Expressing tangent squared in terms of secant squared.                                     | $\tan^2\theta = \sec^2\theta - 1$                                                                |
| 15   | <b>Derived Identity 2b:</b><br>The difference between secant squared and tangent squared.                                 | $\sec^2\theta - \tan^2\theta = 1$                                                                |
| 16   | <b>Fundamental Identity 3:</b><br>The relationship between cotangent and cosecant.                                        | $1 + \cot^2\theta = \csc^2\theta$                                                                |
| 17   | <b>Derived Identity 3a:</b>                                                                                               | $\cot^2\theta = \csc^2\theta - 1$                                                                |

| S.No | Important Formulas / Hints / Points                                                                                             | Formulae                          |
|------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
|      | Expressing cotangent squared in terms of cosecant squared.                                                                      |                                   |
| 18   | <b>Derived Identity 3b:</b><br>The difference between cosecant squared and cotangent squared.                                   | $\csc^2\theta - \cot^2\theta = 1$ |
| 19   | <b>Line of Sight:</b><br>The line drawn from the eye of an observer to the point in the object viewed by the observer.          | Line of sight                     |
| 20   | <b>Angle of Elevation:</b><br>The angle formed by the line of sight with the horizontal when it is above the horizontal level.  | Angle of Elevation                |
| 21   | <b>Angle of Depression:</b><br>The angle formed by the line of sight with the horizontal when it is below the horizontal level. | Angle of Depression               |

### TRIGONOMETRIC RATIO TABLE

| $\theta$     | $0^\circ$ | $30^\circ$           | $45^\circ$           | $60^\circ$           | $90^\circ$                        |
|--------------|-----------|----------------------|----------------------|----------------------|-----------------------------------|
| $\sin\theta$ | 0         | $\frac{1}{2}$        | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | 1                                 |
| $\cos\theta$ | 1         | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$        | 0                                 |
| $\tan\theta$ | 0         | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           | $\infty \left(\frac{1}{0}\right)$ |

## CHAPTER 7: MENSURATION - POINTS TO REMEMBER

| S.No | Important Formulas / Hints / Points                                                                                       | Formulae                           |
|------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 1    | <b>Cuboid - Total Surface Area:</b><br>Where $l$ , $b$ , and $h$ are length, breadth, and height respectively.            | $2(lb + bh + lh)$                  |
| 2    | <b>Cuboid - Volume:</b>                                                                                                   | $l \times b \times h$              |
| 3    | <b>Cube - Total Surface Area:</b> Where $a$ is the side of the cube.                                                      | $6a^2$                             |
| 4    | <b>Cube - Volume:</b>                                                                                                     | $a^3$                              |
| 5    | <b>Right Circular Cylinder - Curved Surface Area (C.S.A):</b><br>Where $r$ is the radius and $h$ is the height.           | $2\pi rh$                          |
| 6    | <b>Right Circular Cylinder - Total Surface Area (T.S.A):</b><br>Sum of C.S.A and area of top and bottom circular regions. | $2\pi r(h + r)$                    |
| 7    | <b>Right Circular Cylinder - Volume:</b>                                                                                  | $\pi r^2 h$                        |
| 8    | <b>Hollow Cylinder - Curved Surface Area:</b><br>Where $R$ and $r$ are outer and inner radii.                             | $2\pi(R + r)h$                     |
| 9    | <b>Hollow Cylinder - Total Surface Area:</b>                                                                              | $2\pi(R + r)(R - r + h)$           |
| 10   | <b>Hollow Cylinder - Volume:</b>                                                                                          | $\pi(R^2 - r^2)h$                  |
| 11   | <b>Right Circular Cone - Curved Surface Area (C.S.A):</b><br>Where $l$ is the slant height.                               | $\pi rl$                           |
| 12   | <b>Right Circular Cone - Slant Height (<math>l</math>):</b><br>Relationship between slant height, radius, and height.     | $l = \sqrt{r^2 + h^2}$             |
| 13   | <b>Right Circular Cone - Total Surface Area (T.S.A):</b>                                                                  | $\pi r(l + r)$                     |
| 14   | <b>Right Circular Cone - Volume:</b>                                                                                      | $\frac{1}{3}\pi r^2 h$             |
| 15   | <b>Frustum of a Cone - Slant Height (<math>l</math>):</b><br>Where $R$ is top radius and $r$ is bottom radius.            | $l = \sqrt{h^2 + (R - r)^2}$       |
| 16   | <b>Frustum of a Cone - Curved Surface Area:</b>                                                                           | $\pi(R + r)l$                      |
| 17   | <b>Frustum of a Cone - Total Surface Area:</b>                                                                            | $\pi(R + r)l + \pi R^2 + \pi r^2$  |
| 18   | <b>Frustum of a Cone - Volume:</b>                                                                                        | $\frac{1}{3}\pi h(R^2 + r^2 + Rr)$ |
| 19   | <b>Sphere - Surface Area:</b> Where $r$ is the radius.                                                                    | $4\pi r^2$                         |
| 20   | <b>Sphere - Volume:</b>                                                                                                   | $\frac{4}{3}\pi r^3$               |
| 21   | <b>Hollow Sphere - Volume:</b><br>Where $R$ is outer radius and $r$ is inner radius.                                      | $\frac{4}{3}\pi(R^3 - r^3)$        |
| 22   | <b>Hemisphere - Curved Surface Area (C.S.A):</b>                                                                          | $2\pi r^2$                         |
| 23   | <b>Hemisphere - Total Surface Area (T.S.A):</b>                                                                           | $3\pi r^2$                         |

| S.No | Important Formulas / Hints / Points      | Formulae                    |
|------|------------------------------------------|-----------------------------|
| 24   | Hemisphere - Volume:                     | $\frac{2}{3}\pi r^3$        |
| 25   | Hollow Hemisphere - Curved Surface Area: | $2\pi(R^2 + r^2)$           |
| 26   | Hollow Hemisphere - Total Surface Area:  | $\pi(3R^2 + r^2)$           |
| 27   | Hollow Hemisphere - Volume:              | $\frac{2}{3}\pi(R^3 - r^3)$ |

## CHAPTER 8: STATISTICS AND PROBABILITY - POINTS TO REMEMBER

| S.No | Important Formulas / Hints / Points                                                                                                                                                              | Formulae                                                                   |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| 1    | <b>Range:</b><br>The difference between the largest value ( $L$ ) and the smallest value ( $S$ ) in a given data set.                                                                            | $R = L - S$                                                                |
| 2    | <b>Coefficient of Range:</b><br>A relative measure of dispersion based on the range.                                                                                                             | $C.V = \frac{L - S}{L + S}$                                                |
| 3    | <b>Variance (Ungrouped Data):</b><br>The mean of the squares of the deviations from the mean. Denoted by $\sigma^2$ .                                                                            | $\sigma^2 = \frac{\sum(x_i - \bar{x})^2}{n}$                               |
| 4    | <b>Standard Deviation (Ungrouped Data):</b> The positive square root of the variance. It is the most widely used measure of dispersion.                                                          | $\sigma = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n}}$                          |
| 5    | <b>Standard Deviation (Direct Method):</b><br>Formula to calculate standard deviation directly without calculating deviations from the mean.                                                     | $\sigma = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2}$ |
| 6    | <b>Standard Deviation (Assumed Mean Method):</b><br>Used when data values are large. Here $d_i = x_i - A$ , where $A$ is the assumed mean.                                                       | $\sigma = \sqrt{\frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n}\right)^2}$ |
| 7    | <b>Standard Deviation of first 'n' natural numbers:</b> A direct formula for the standard deviation of 1,2,3, ..., n.                                                                            | $\sigma = \sqrt{\frac{n^2 - 1}{12}}$                                       |
| 8    | <b>Properties of Standard Deviation (Addition/Subtraction):</b> If a constant $k$ is added to or subtracted from each data value, the standard deviation remains unchanged.                      | New $\sigma$ = Old $\sigma$                                                |
| 9    | <b>Properties of Standard Deviation (Multiplication/Division):</b> If each data value is multiplied or divided by a constant $k$ , the standard deviation is also multiplied or divided by $k$ . | New $\sigma =  k  \times$ Old $\sigma$                                     |
| 10   | <b>Coefficient of Variation (C.V.):</b><br>The standard deviation expressed as a percentage of the mean. It is used to compare the consistency of two or more data sets.                         | $C.V. = \frac{\sigma}{\bar{x}} \times 100\%$                               |
| 11   | <b>Consistency and C.V.:</b><br>The data set with a lesser Coefficient of Variation is more consistent or stable. The data set with a higher C.V. is                                             | Lesser C.V. $\Rightarrow$ More consistent                                  |

| S.No | Important Formulas / Hints / Points                                                                                                                                        | Formulae                                                                                                                                                                                                                                                                                   |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | more variable.                                                                                                                                                             |                                                                                                                                                                                                                                                                                            |
| 12   | <b>Probability of an Event:</b><br>The ratio of the number of favorable outcomes for event $A$ to the total number of possible outcomes in the sample space $S$ .          | $P(A) = \frac{n(A)}{n(S)}$                                                                                                                                                                                                                                                                 |
| 13   | <b>Limits of Probability:</b><br>The probability of any event $A$ always lies between 0 and 1 (inclusive).                                                                 | $0 \leq P(A) \leq 1$                                                                                                                                                                                                                                                                       |
| 14   | <b>Sure Event:</b><br>An event that is certain to happen has a probability of 1.                                                                                           | $P(S) = 1$                                                                                                                                                                                                                                                                                 |
|      | Sample Space: <b>1 Coin</b>                                                                                                                                                | $S = \{H, T\} \Rightarrow n(S) = 2$                                                                                                                                                                                                                                                        |
|      | Sample Space: <b>2 Coins</b>                                                                                                                                               | $S = \{HH, HT, TH, TT\} \Rightarrow n(S) = 4$                                                                                                                                                                                                                                              |
|      | Sample Space: <b>3 Coins</b>                                                                                                                                               | $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\} \Rightarrow n(S) = 8$                                                                                                                                                                                                                      |
|      | Sample Space: <b>1 Die</b>                                                                                                                                                 | $S = \{1, 2, 3, 4, 5, 6\} \Rightarrow n(S) = 6$                                                                                                                                                                                                                                            |
|      | Sample Space: <b>2 Dice</b>                                                                                                                                                | $S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\} \Rightarrow n(S) = 36$ |
| 15   | <b>Impossible Event:</b><br>An event that cannot happen has a probability of 0.                                                                                            | $P(\phi) = 0$                                                                                                                                                                                                                                                                              |
| 16   | <b>Complementary Event:</b><br>The probability of an event $A$ not happening, denoted by $\bar{A}$ or $A'$ .                                                               | $P(\bar{A}) = 1 - P(A) \text{ or } P(A) + P(\bar{A}) = 1$                                                                                                                                                                                                                                  |
| 17   | <b>Addition Theorem of Probability:</b><br>For any two events $A$ and $B$ , the probability of occurrence of at least one of them.                                         | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$                                                                                                                                                                                                                                                  |
| 18   | <b>Mutually Exclusive Events:</b><br>If two events $A$ and $B$ cannot occur simultaneously, then their intersection is null ( $A \cap B = \phi$ ), and $P(A \cap B) = 0$ . | $P(A \cup B) = P(A) + P(B)$                                                                                                                                                                                                                                                                |
| 19   | <b>Addition Theorem for Three Events:</b><br>For any three events $A$ , $B$ , and $C$ .                                                                                    | $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$                                                                                                                                                                                   |